

SECTION 1 – SYSTEM CONFIGURATION

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
1-1	INTRODUCTION	1-3
1-2	UPGRADE TO SIGNA MR/i FROM SIGNA HORIZON LX WITH LCC MAGNET	1-5
1-3	SIGNA MR/i SYSTEM OPTIONS	1-6
1-4	UPGRADE TO SIGNA CV/i FROM SIGNA CVMR WITH LCC MAGNET	1-7
1-5	TRANSPORTABLE/RELOCATABLE UPGRADE	1-9
1-6	TRUE MOBILE UPGRADE	1-9
1-7	VAN MANUFACTURERS CONTACT INFORMATION	1-10

1-1 INTRODUCTION

The Upgrade configurations documented throughout this publication are:

<u>CURRENT SYSTEM CONFIGURATION</u>	➤	<u>UPGRADED SYSTEM CONFIGURATION</u>
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Signa Horizon LX with LCC Magnet	➤	Signa MR/i with LCC Magnet & WideOpen Enclosure
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Signa CVMR with LCC Magnet	➤	Signa CV/i with LCC Magnet & WideOpen Enclosure
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Note

The WideOpen Enclosure Upgrade is only available for the LCC Magnet.

Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in SECTION 6 INTERCONNECT DATA.

- SECTION 1-2, UPGRADE TO SIGNA MR/i FROM SIGNA HORIZON LX WITH LCC MAGNET
 - Fixed, Transportable, or Relocatable sites
 - True Mobile sites
- SECTION 1-3, SIGNA MR/i SYSTEM OPTIONS
- SECTION 1-4, UPGRADES TO SIGNA CV/i FROM 1.5T SIGNA CVMR WITH LCC MAGNET
 - Fixed, Transportable, or Relocatable sites

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Pre-Installation document uses the technical document names.

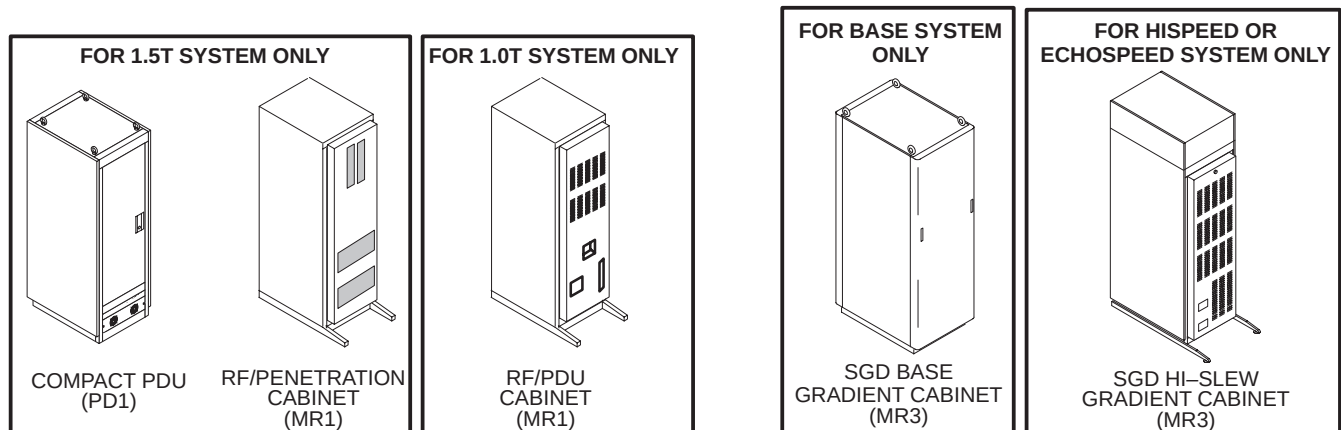
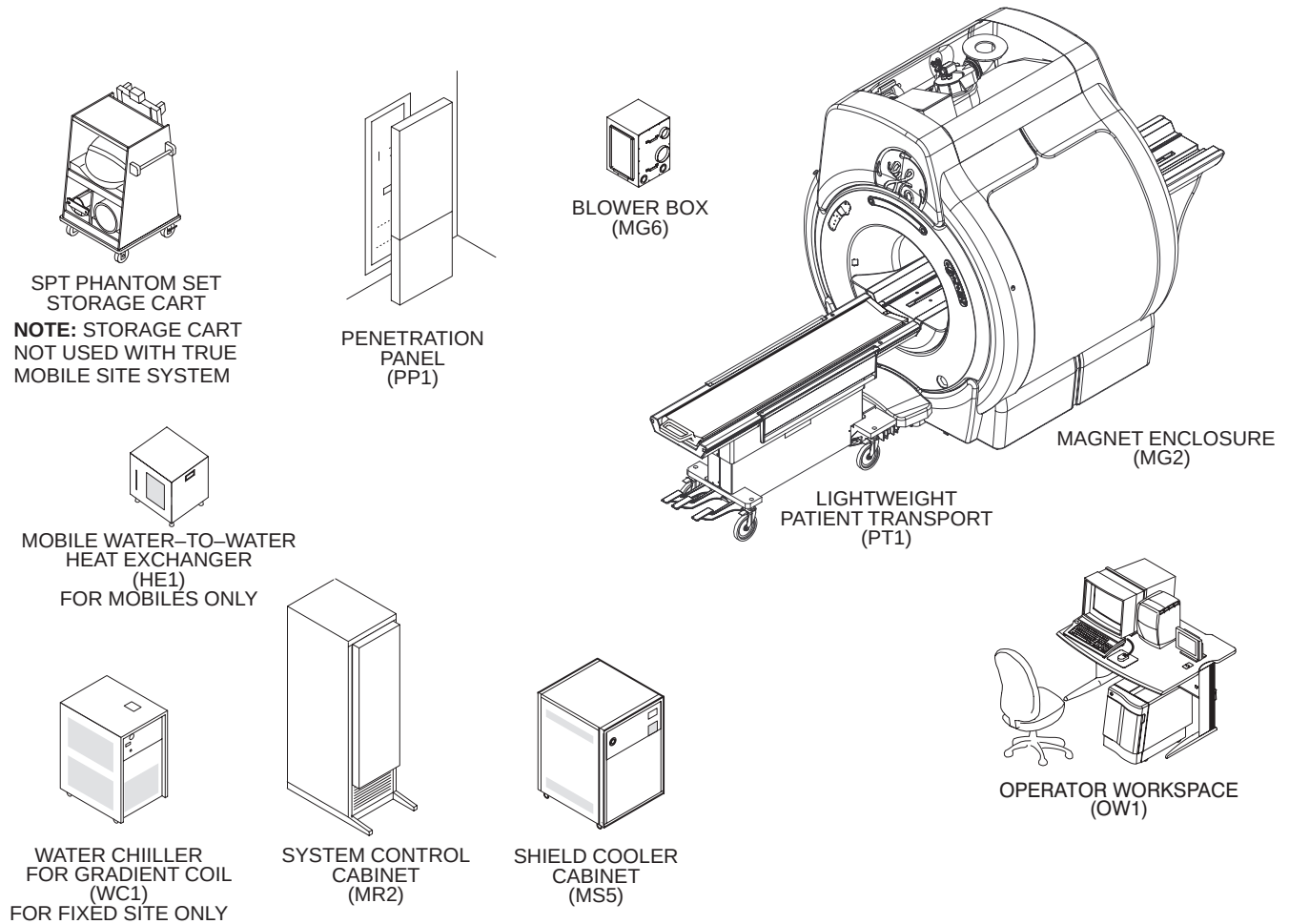
SALES LITERATURE NAME	= TECHNICAL DOCUMENT NAME
CX 100 Magnet with K4 Technology (CXK4)	= 1.0T LCC (Active Shield) Magnet
CX 150 Magnet with K4 Technology (CXK4)	= 1.5T LCC (Active Shield) Magnet
SX Gradient Subsystem	= SGD (Scaleable Gradient Driver)

Illustration 1-1 shows the system's major equipment after a Signa MR/i Upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final product configuration. The following sub-sections provide an overview of the various Signa Horizon LX with SGD upgrade hardware.

Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

1-1 INTRODUCTION (Continued)



SIGNA MR/i SYSTEM MAJOR EQUIPMENT

ILLUSTRATION 1-1

1-2 UPGRADE TO SIGNA MR/i FROM SIGNA HORIZON LX WITH LCC MAGNET

The following tables lists the Signa MR/i upgrade catalogs for Signa Horizon LX system with an LCC Magnet. Illustration 1-2 shows major equipment impacted (added, replaced, and modified) by the listed catalogs.

TABLE 1-1
SIGNA MR/i UPGRADE SITE COLLECTOR KIT (SELECT ONE)

CATALOG	DESCRIPTION
M1090YL	1.5T/1.0T LCC Magnet WideOpen Enclosure Upgrade for fixed site, Transportable, or Relocatable ●Bridge for WideOpen Enclosure with BRM Gradient Coil ●New Rear Pedstal and new Rear Pedestal covers (electronic parts are transferred from old Rear Pedestal) ●Remote PAC Interface Assembly (mounts to new enclosure) ●Fixed site enclosure trim and removable cover pieces. ●New fixed site Bore Light Assembly ●Dock Cover Assembly ●PDU modification to power Water Chiller for Gradient Coil Cooling
M1090YN	1.5T/1.0T LCC Magnet WideOpen Enclosure Upgrade for True Mobile ●Bridge for WideOpen Enclosure with BRM Gradient Coil ●New Rear Pedstal and new Rear Pedestal covers (electronic parts are transferred from old Rear Pedestal) ●New Dock-to-magnet mounting bracket ●Remote PAC Interface Assembly (mounts to new enclosure) ●True Moible system removable cover pieces. ●New Mobile Bore Light Assembly ●Dock Cover Assembly

TABLE 1-2
SIGNA MR/i UPGRADE MAGNET ENCLOSURE COLLECTOR (SELECT ONE)

CATALOG	DESCRIPTION
M1090YW	Fixed, Transportable, or Relocatable site LCC Magnet WideOpen Enclosure Upgrade
M1090YY	True Mobile LCC Magnet WideOpen Enclosure Upgrade

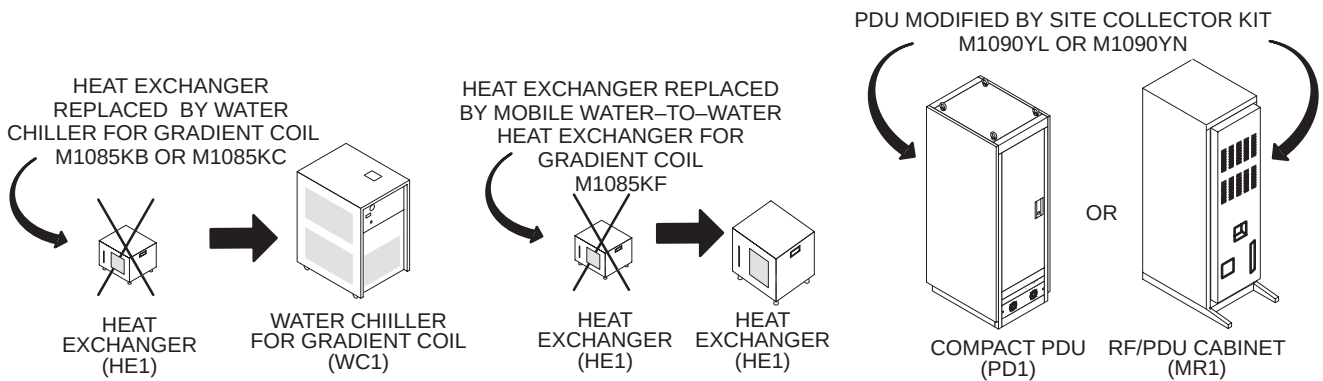
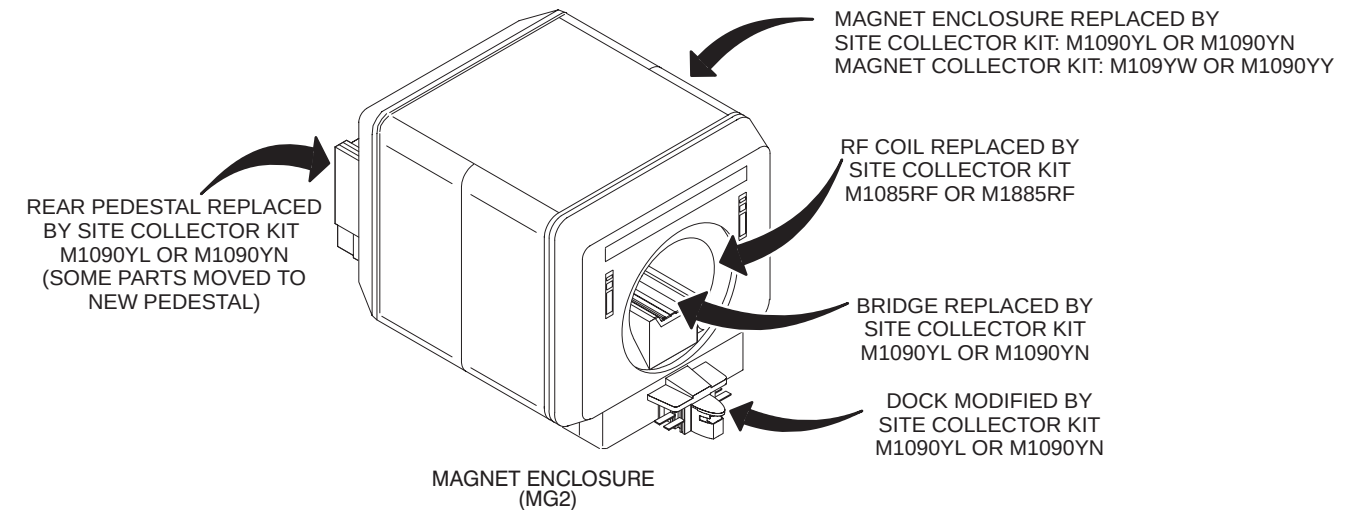
TABLE 1-3
EQUIPMENT FOR GRADIENT COIL COOLING (SELECT ONE)

CATALOG	DESCRIPTION
M1085KC	●60 Hz Water-to-Air Chiller for Gradient Coil Cooling <i>See Note 1</i>
M1085KB	●50 Hz Water-to-Air Chiller for Gradient Coil Cooling <i>See Note 1</i>
M1085KF	●50/60 Hz Water-to-Water Heat Exchanger for Mobile systems Gradient Coil Cooling <i>See Note 2</i>
Note 1 Fixed Site must order Water Chiller for their power frequency (60 or 50 Hz).	
Note 2 Mobile (True, Transportable, Relocatable) systems require this heat exchanger. The Mobile van will provide chilled water for the heat exchanger operation.	

TABLE 1-4
WIDEOPEN ENCLOSURE COMPATIBLE RF COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085RF	1.5T RF Coil compatible with WideOpen Enclosure
M1885RF	1.0T RF Coil compatible with WideOpen Enclosure

1-2 UPGRADE TO SIGNA MR/i FROM SIGNA HORIZON LX WITH LCC MAGNET (Continued)



SIGNA HORIZON LX SYSTEM WITH LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-2

1-3 SIGNA MR/i SYSTEM OPTIONS

There are no system options specific to the Signa MR/i Upgrade. Refer to Direction 2223170 Signa MR/i Pre-Installation for typical system options information.

1-4 UPGRADE TO SIGNA CV/i FROM SIGNA CVMR WITH LCC MAGNET

The following tables lists the Signa CV/i upgrade catalogs for Signa CVMR system with an LCC Magnet in a fixed site or Transportable/Relocatable Mobile configuration. Illustration 1-2 shows major equipment impacted (added, replaced, and modified) by the listed catalogs.

Note

Signa CV/i upgrade is not available for True Mobiles or 1.0T system configurations.

TABLE 1-5
SIGNA CV/i UPGRADE SITE COLLECTOR KIT

CATALOG	DESCRIPTION
M1091YM	1.5T LCC Magnet WideOpen CV/i Enclosure Upgrade for fixed site, Transportable, or Relocatable ● Bridge for WideOpen Enclosure with CRM Gradient Coil ● New Rear Pedestal and new Rear Pedestal covers (electronic parts are transferred from old Rear Pedestal) ● Remote PAC Interface Assembly (mounts to new enclosure) ● Fixed site enclosure trim and removable cover pieces. ● New fixed site Bore Light Assembly ● Dock Cover Assembly ● PDU modification to power Water Chiller for Gradient Coil Cooling

TABLE 1-6
SIGNA CV/i UPGRADE MAGNET COLLECTOR

CATALOG	DESCRIPTION
M1090YP	Fixed, Transportable, or Relocatable site LCC Magnet WideOpen Enclosure Upgrade

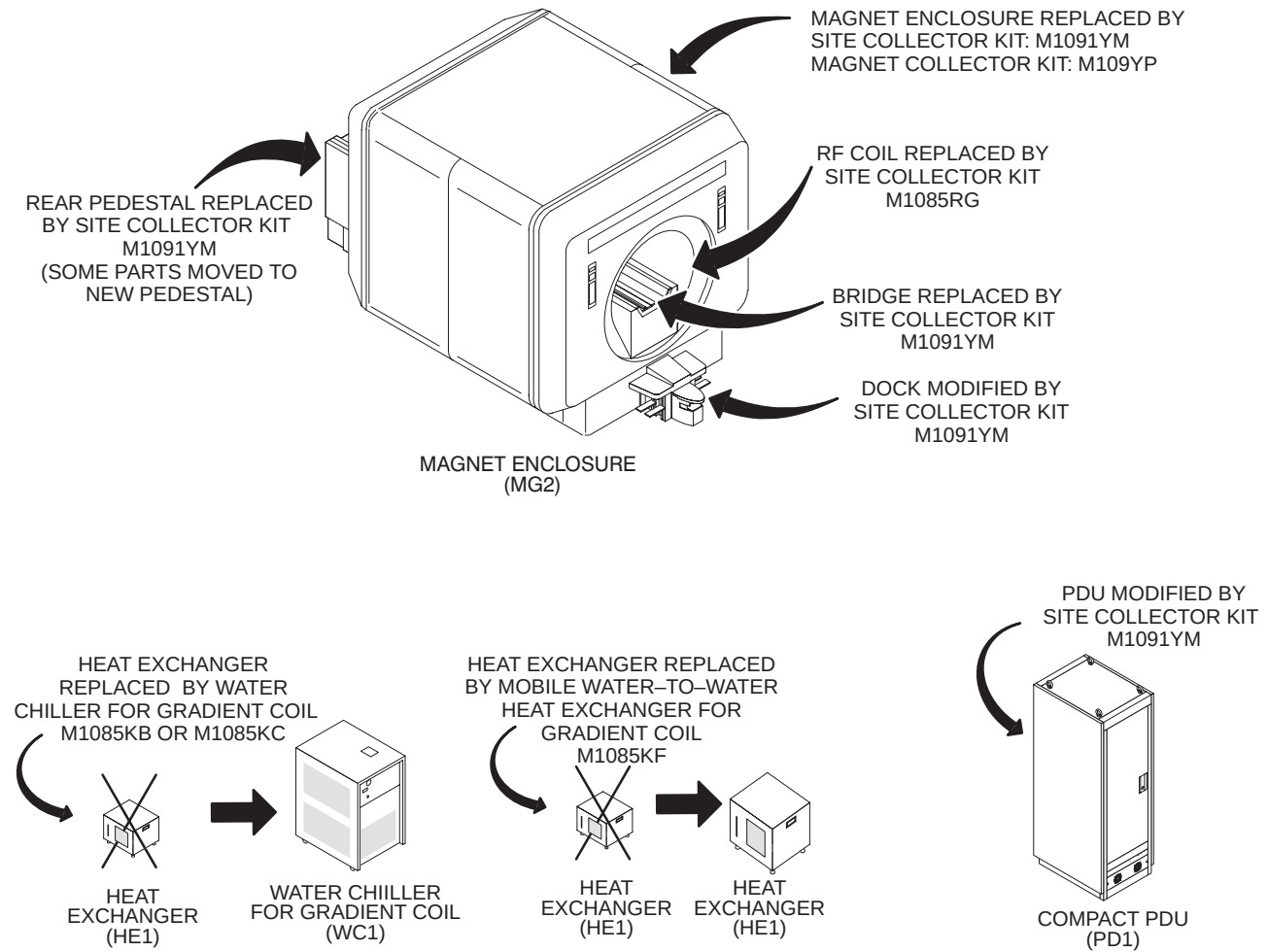
TABLE 1-7
EQUIPMENT FOR GRADIENT COIL COOLING (SELECT ONE)

CATALOG	DESCRIPTION
M1085KC	● 60 Hz Water-to-Air Chiller for Gradient Coil Cooling <i>See Note 1</i>
M1085KB	● 50 Hz Water-to-Air Chiller for Gradient Coil Cooling <i>See Note 1</i>
M1085KF	● 50/60 Hz Water-to-Water Heat Exchanger for Mobile systems Gradient Coil Cooling <i>See Note 2</i>
Note 1 Fixed Site must order Water Chiller for their power frequency (60 or 50 Hz). Transportable and Relocatable systems do not require this chiller, Gradient Coil chilled water will be provided by Van chilled water system.	
Note 2 Transportable and Relocatable systems require this heat exchanger. The Transportable/Relocatable van will provide chilled water for the heat exchanger operation.	

TABLE 1-8
WIDEOPEN ENCLOSURE COMPATIBLE SIGNA CV/i RF COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085RG	● Signa CV/i RF Coil compatible with WideOpen Enclosure

1-4 UPGRADE TO SIGNA CV/i FROM SIGNA CVMR WITH LCC MAGNET (Continued)



SIGNA CVMR SYSTEM WITH LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-3

1-5 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable Signa Horizon LX system with LCC Magnet or Signa Horizon CVMR system with LCC Magnet upgrade catalogs are listed in the previous upgrade sections.

In addition to the system, the van must also be upgraded to accommodate the Signa MR/i and CV/i upgrades. The van/system upgrades can typically be performed at the customer site. It is customer responsibility to contact van manufacturer to confirm viability to perform upgrade at customer site.

- Customer must contact Van Manufacturer to verify upgrade can be performed at customer site. Refer to Section 1-7, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- System check and complete calibrations are performed by GE Field Service.

1-6 TRUE MOBILE UPGRADE

True Mobile Signa Horizon LX system with LCC Magnet upgrade catalogs are listed in the Section 1-2 UPGRADE TO SIGNA MR/i FROM SIGNA HORIZON LX WITH LCC MAGNET.

In addition to the system, the van must also be upgraded to accommodate the Signa MR/i upgrades. The True Mobile van/system upgrade are performed at the Van Manufacturer facility.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility. Refer to Section 1-7, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility.
- System check and complete calibrations are performed by MR Manufacturing.

1-7 VAN MANUFACTURERS CONTACT INFORMATION

- Calumet Coach Company
2150 East Dolton Road, PO Box 1369
Calumet City, IL 60409-1411
Telephone: 708-868-5070
FAX: 708-868-5101
- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
Telephone: 513-752-9000
FAX: 513-752-4983
- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
Telephone: 414-367-7700
FAX: 414-367-7744
- AK Associates Inc.
16745 South Lathrop Avenue
Harvey, IL 60426
USA
Telephone: 708-596-5066
FAX: 708-596-2480